RS4 — The effective action of Cornwall, Jackiw and Tomboulis along the deflation clock

PROJECT spacetimes, Revision_Deflation. Supplement to both manuscripts of the revision: the paired state, matter, temperature and the vacuum of the two models as the couplings grow

Prepared for Patrick Lee Nash, Ph.D.

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0. Purpose, and summary of the results

The supplement S4 of the main project derives the properties of the two models from one function, the effective action for composite operators of Cornwall, Jackiw and Tomboulis (CJT): the gap equation of the paired state is its stationarity condition, the condensation energy is its value, and the same action gives the fermion mass in matter, its melting at high temperature, and the stability of the vacuum. S4 does this for one value of the coupling at a time.

The supplement RS1 of the revision shows that on the deflating spacetimes the models of the main project hold unchanged for the weighted fields, with a coupling that grows with the ordinary time: $g_{VS} \propto e^{\tau_d/2}$, where $\tau_d = \mathrm{gsum}\,(x4 - x4_{\mathrm{ref}})$ is the deflation clock. This supplement therefore repeats every calculation of S4 *along the clock*: at a sequence of moments, each with the coupling of that moment. The reader needs S4 for the method and RS1, Sec. 1, for the clock. The results:

1. **The paired state** (Sec. 2). At every moment the solution of the gap equation is a minimum of the CJT potential. The condensation energy deepens monotonically, in weak coupling with twice the logarithmic slope of the gap. The pairs shrink: the coherence length falls from 25 to 1.0 in units of $1/k_F$ along the history of one particle, which carries the state from overlapping pairs to the edge of bound pairs. The universal ratios of the weak-coupling theory hold until the gap approaches a fifth of the Fermi energy.
2. **Matter** (Sec. 3). Matter of any fixed density loses its fermion mass at a definite moment, the denser the earlier. Once $C > 3.7011$ matter is self-bound, and it becomes more strongly bound as the deflation proceeds.
3. **Temperature** (Sec. 4). The temperature at which the fermion mass melts falls as the coupling grows: matter of a fixed temperature melts at a definite moment.
4. **The vacuum** (Sec. 5). The field beyond which the one-loop potential of the Vorton-Scalar lies below its vacuum value moves in towards the vacuum value as the coupling grows: the barrier that protects the vacuum weakens.

Every one of these is a statement about a *sequence of equilibrium states*. Whether a system can follow the sequence is examined in RS1, Sec. 7.

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
# Sec. 2 and the last part of Sec. 3 (about ten seconds)
./code/deflation-rs/target/release/deflation paired
# Secs. 3 to 5 (about a minute and a half)
./code/deflation-rs/target/release/deflation history
```

```
python3 code/scripts/make_figures.py cjt matter
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py docs_src/RS4_effective_action_along_the_clock.src.md
bash code/scripts/build_docs.sh supplement/RS4_effective_action_along_the_clock.md
```

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1. The method, and why it carries over

S4, Secs. 1 and 2, defines the CJT action $\Gamma[\phi, G]$ of the two models: a function of the expectation value ϕ of the Vorton-Scalar and of the full propagators, whose two-particle-irreducible part Γ_2 is truncated at two loops. Its stationarity conditions are the equation for the fermion mass in matter and the gap equation. On the deflating spacetimes, and for fields that do not depend on the extra coordinates, nothing in this construction changes, for two reasons established in RS1:

- the weighted fermion field obeys the flat equations, so its propagator is the flat one;
- the interaction between the fermions is, at each moment, the exchange of the two modes of the scalar with the coupling of that moment, and the static potential has the unshifted range (RS1, Sec. 2.2).

The one new term, the shift $-\mathrm{gsum}^2/4$ of the square of the boson masses (RS1, Eq. (1.2)), is smaller than every other scale of the calculation by the square of the ratio of the rate of deflation to a particle mass. For the largest rate that observation allows (supplement RS3) and a mass of 1 keV that ratio is below 10^{-36}. The term is neglected here; this is the only approximation that this supplement makes beyond those of S4.

The test of the method. S4, Sec. 1.7, tests the truncation of Γ_2 on a model in zero dimensions whose exact solution is known. The test does not involve the spacetime. It is executed again by the program of the revision: the two-loop truncation has a solution for every coupling from 0.01 to 30 and stays within 10.7 per cent of the exact propagator (check `def-P5`).

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2. The paired state along the clock

2.1 A minimum at every moment

The CJT potential $V[\Delta]$ of the paired state is the effective action per unit volume, measured from the unpaired state, as a functional of the gap function $\Delta(k)$ (S4, Sec. 3.1). The program evaluates it at 13 moments for two particles — that is, for two choices of the mass m_s of the radial mode at the Fermi momentum $k_F = 0.5\,m$ — in the vector channel:

m_s/m, k_F/m	τ_d	g_{VS}	λ	gap$/m$	V/m^4	ratio	curvature	T_c/gap	size $\times k_F$
0.30, 0.50	0.00	1.80	0.216	2.81×10^{-3}	-4.46×10^{-7}	1.000	0.21	0.567	25.4
0.30, 0.50	0.21	2.00	0.266	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.26	0.566	10.3
0.30, 0.50	0.40	2.20	0.322	0.0135	-1.03×10^{-5}	0.992	0.30	0.564	5.3
0.30, 0.50	0.58	2.40	0.383	0.0230	-2.93×10^{-5}	0.981	0.34	0.559	3.1
0.30, 0.50	0.74	2.60	0.450	0.0354	-6.81×10^{-5}	0.961	0.37	0.552	2.0
0.30, 0.50	0.88	2.80	0.522	0.0511	-1.38×10^{-4}	0.932	0.38	0.542	1.4

m_s/m, k_F/m	τ_d	g_{VS}	λ	gap$/m$	V/m^4	ratio	curvature	T_c/gap	size $\times k_F$
0.30, 0.50	1.02	3.00	0.599	0.0707	-2.53×10^{-4}	0.895	0.38	0.530	1.0
1.0, 0.50	0.00	2.40	0.133	4.25×10^{-4}	-1.02×10^{-8}	1.000	0.13	0.567	167.6
1.0, 0.50	0.16	2.60	0.156	1.32×10^{-3}	-9.88×10^{-8}	1.000	0.15	0.567	53.9
1.0, 0.50	0.31	2.80	0.181	3.28×10^{-3}	-6.08×10^{-7}	0.999	0.17	0.567	21.7
1.0, 0.50	0.45	3.00	0.208	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.19	0.566	10.3
1.0, 0.50	0.58	3.20	0.237	0.0128	-9.25×10^{-6}	0.994	0.21	0.564	5.6
1.0, 0.50	0.70	3.40	0.267	0.0218	-2.65×10^{-5}	0.987	0.23	0.561	3.3

(The columns: the pairing strength λ; the gap at the Fermi momentum; the potential V at the solution, which is the condensation energy; its ratio to the weak-coupling value $-\nu N(0)\,\text{gap}^2/4$, with $\nu = 8$ internal states and the density of states $N(0)$; the curvature, that is, the smallest eigenvalue of the matrix of second derivatives of the potential; the ratio of the critical temperature to the gap; and the size of the pair — its coherence length — in units of $1/k_F$.) The results (check `def-P1`):

1. *The solution of the gap equation is a minimum.* Newton's method applied to the potential converges to the gap function that the iteration of the gap equation finds, to 7.7×10^{-12}. Along the ray through the solution the potential is lowest at the solution, to 4.6×10^{-7}. The smallest eigenvalue of the second variation is positive at every moment.
2. *Two routes to the critical temperature agree.* The temperature at which the second variation of the potential at $\Delta = 0$ loses its positivity equals the temperature at which the linearized gap equation has the eigenvalue 1, to 2.6×10^{-12}.
3. *The condensation energy deepens.* It is within 15 per cent of the weak-coupling value wherever the gap is below $0.1\,E_F$, and in weak coupling its logarithmic slope along the clock is twice that of the gap, to 0.1 per cent: by RS1, Eq. (3.2),
$$\frac{d\ln|V|}{d\tau_d} = \frac{2}{\lambda}.$$
4. *The pairs shrink.* The coherence length of the first particle falls from 25 to 1.0 in units of $1/k_F$. At its last five moments the gap exceeds $0.1\,E_F$, and at the last the pair is as small as the distance between the particles. **The deflation carries the paired state from overlapping pairs towards bound pairs.** On flat spacetime this crossover is reached by choosing a larger coupling; here it is reached by waiting.

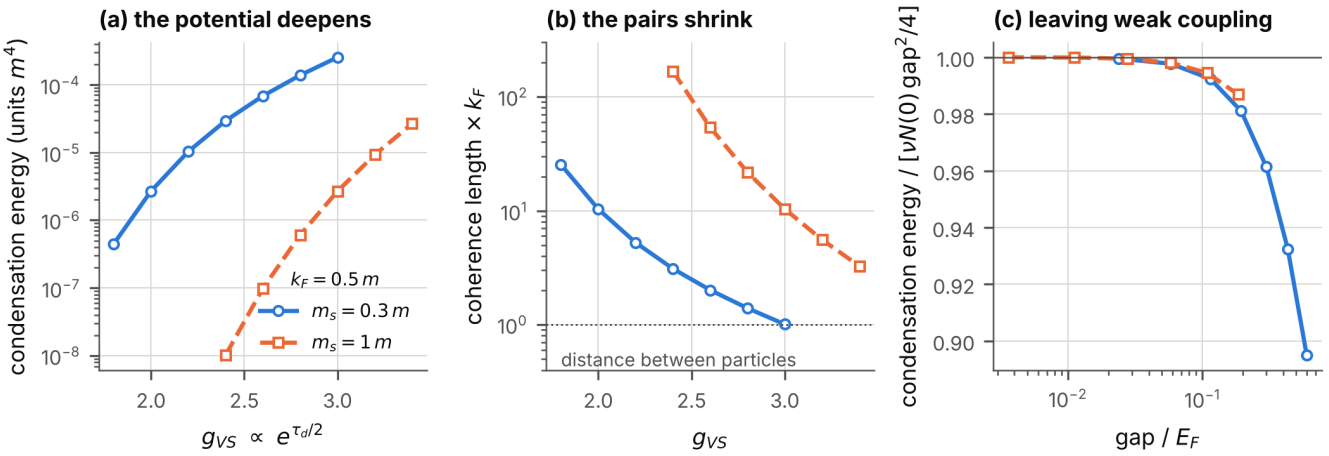


Figure 1: The paired state along the clock, from the CJT potential, for two particles. (a) The condensation energy against the coupling of the moment. (b) The coherence length of the pair in units of $1/k_F$. (c) The condensation energy divided by its weak-coupling value, against the gap in units of the Fermi kinetic energy: the departure from weak coupling. Data: `data/cjt_history.csv`; program: `deflation paired`.

2.2 Thermodynamics

At three moments the program follows the paired state of the second particle from zero temperature to the critical temperature (check `def-P2`):

τ_d	g_{VS}	λ	T_c/m	T_c/E_F	gap(0)/T_c	jump of the heat capacity
0.000	2.60	0.156	7.49×10^{-4}	0.0063	1.7640	1.4266
0.286	3.00	0.208	3.90×10^{-3}	0.0330	1.7668	1.4331
0.537	3.40	0.267	0.0122	0.1036	1.7828	1.4742

While the critical temperature rises by a factor of sixteen, from 6.34×10^{-3} to 1.04×10^{-1} of the Fermi energy, the ratio of the gap at zero temperature to the critical temperature stays within 0.019 of the weak-coupling value 1.7639, and the relative jump of the heat capacity at T_c within 0.048 of 1.4261. Both ratios are universal while the coupling is weak; they begin to move at the last moment, where the gap approaches $0.2\,E_F$.

2.3 The Ginzburg–Landau energy

Near the critical temperature the potential reduces to the Ginzburg–Landau energy of S1, Sec. 6.2, with three coefficients: the quadratic coefficient $a(T)$, the quartic coefficient b and the gradient coefficient K_{GL}. The program extracts them from the CJT potential at the same three moments and compares them with the weak-coupling formulas (check `def-P3`):

τ_d	g_{VS}	T_c/m	ratio for the slope of $a(T)$	ratio for b	ratio for K_{GL}	coherence length × m	coherence length × T_c/v_F
0.000	2.60	7.49×10^{-4}	1.0000	1.0001	1.0005	79.6	0.1333
0.286	3.00	3.90×10^{-3}	0.9993	0.9949	1.0101	15.4	0.1340
0.537	3.40	0.0122	0.9927	0.9641	1.0793	5.1	0.1390

(The three columns of ratios are the computed coefficient divided by its weak-coupling formula.) While T_c is below $0.05\,E_F$ the coefficients agree with the formulas to 1.0 per cent; at the third moment, $T_c = 0.10\,E_F$, they differ by up to 7.9 per cent. The consequences of the formulas along the clock are:

- the quadratic coefficient keeps its slope with respect to $\ln T$, which contains the density of states only;
- the quartic and the gradient coefficients fall as $1/T_c^2$, hence as the inverse square of the gap;
- the coherence length of the Ginzburg–Landau energy is $0.1333\,v_F/T_c$ in weak coupling (computed: 0.1333, 0.1340, 0.1390 at the three moments) and shrinks from 80 to 5.1 in units of $1/m$. **The cores of the vortices of the superfluid shrink as the deflation proceeds.**

2.4 The space of the order parameter

The potential as a function of the direction and of the phase of the order parameter Δ^a is discussed in RS1, Sec. 5.2, with its table: every unitary state has the potential -2 in units of $N(0)\,\text{gap}^2$, with four flat and four positive directions; the half-paired state has -1; and the deflating axes are not distinguished from $x0$ (check `def-P4`).

2.5 The bound pair, from the gap equation

In vacuum the linearized gap equation is the equation of a bound state of two fermions. On the deflating spacetime $D_{VS} = \alpha_{VS} m/m_s$ grows as e^{τ_d} and reaches the value at which the equation has the eigenvalue 1 at a definite moment (check `def-P6`):

m_s/m	D_c from the gap equation	D_c from the Schrödinger equation	difference
0.30	1.7347	1.6798	3.3 %
0.10	1.6902	1.6798	0.6 %
0.030	1.6810	1.6798	0.1 %

The gap equation has a relativistic kernel, the Schrödinger equation of RS1, Sec. 4, does not; the two thresholds approach each other as the radial mode becomes light and the binding non-relativistic.

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3. Matter along the clock

3.1 Matter of fixed density becomes abnormal

In matter the expectation value ϕ of the Vorton-Scalar is reduced by the scalar density of the fermions, and with it the fermion mass $m^* = g_{VS}\phi$ in the matter. S4, Sec. 5, shows that the reduction depends on the density and on the single number $C = g_{VS}^2 m^2/m_s^2$, and that above a critical value of C the mass vanishes: the matter is then in the **abnormal** state, a gas of massless fermions in a region without expectation value of the scalar. On the deflating spacetime $C \propto e^{\tau_d}$ grows at fixed density. The critical values (check `def-H5`):

Fermi momentum k_F/m	0.2	0.5	1	2
critical C	151.892	12.971	2.556	0.617
divided by the estimate $\pi^2 m^2/4k_F^2$ of S4	2.46	1.31	1.04	1.00

Every piece of matter becomes abnormal at a definite moment, the denser the earlier. The estimate of the main project is reached at high density, where the transition happens at weak coupling; at low density the critical value is larger.

3.2 Self-bound matter

S4 shows that matter is self-bound — its energy per particle at some density lies below the mass of a free fermion — if $C > 3\pi^2/8 = 3.7011$ without the filled negative-energy states and if $C > 3\pi^2/5 = 5.9218$ with them (check `def-H1`). Along the clock (check `def-P7`; mean field without the negative-energy states, $C = 4\,e^{\tau_d}$):

τ_d	C	energy per particle $-\,m$ (units m)	k_F/m at the minimum	m^*/m there	exchange, radial mode	exchange, phase mode
0.00	4.00	-0.0192	0.981	0.000	0.44 %	0.51 %
0.25	5.14	-0.0787	0.921	0.000	0.47 %	0.54 %
0.50	6.59	-0.1345	0.866	0.000	0.50 %	0.59 %
0.75	8.47	-0.1869	0.813	0.000	0.54 %	0.66 %
1.00	10.87	-0.2362	0.764	0.000	0.59 %	0.73 %
1.25	13.96	-0.2825	0.718	0.000	0.65 %	0.82 %
1.50	17.93	-0.3259	0.674	0.000	0.72 %	0.93 %
1.75	23.02	-0.3668	0.633	0.000	0.80 %	1.06 %
2.00	29.56	-0.4051	0.595	0.000	0.90 %	1.22 %

Once the threshold is passed, the lowest energy per particle falls monotonically, from $-0.0192\,m$ at $C = 4$ to $-0.405\,m$ at $C = 29.6$. At every one of these moments the matter of lowest energy is in the abnormal state, at a Fermi momentum that falls from $0.98\,m$ to $0.59\,m$. The exchange energies of the two modes at that density, which the mean field omits, are 0.44 and 0.51 per cent of the mean-field energy at the first moment and 0.90 and 1.22 per cent at the last (for $m_s/m = 0.3$): the mean field remains a good approximation along the whole history.

Meaning. On a deflating spacetime, a gas of the fermions of the model that is dilute and unbound at early times has, at late times, a state of lower energy: drops of dense abnormal matter, bound by up to 40 per cent of the rest mass at the last moment of the table. Whether and how fast a gas reaches that state is a question of kinetics that this supplement does not answer.

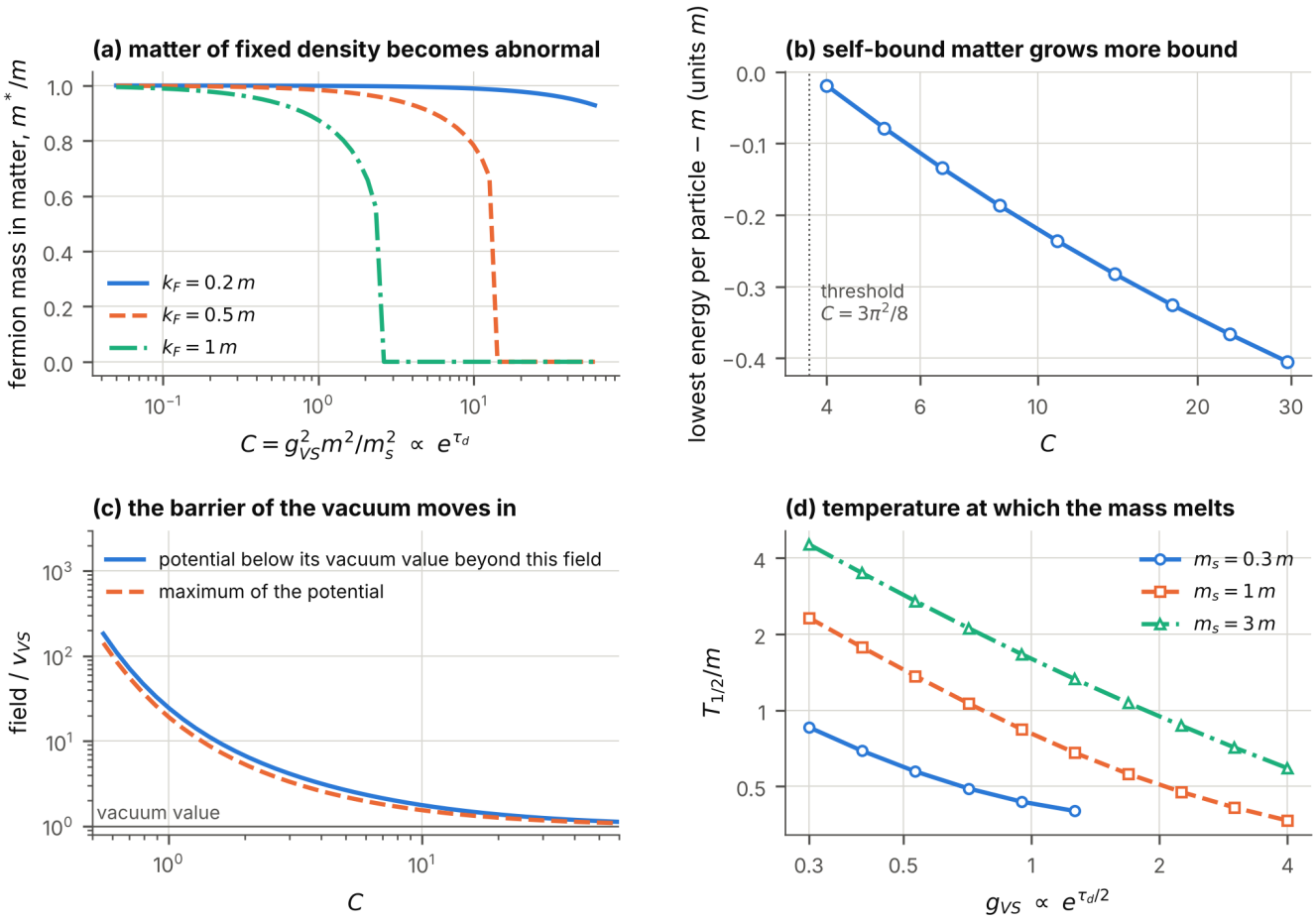


Figure 2: Matter, vacuum and temperature along the clock. (a) The fermion mass in matter of three fixed densities against $C \propto e^{\tau_d}$: it drops to zero at the critical value. (b) The lowest energy per particle of self-bound matter, minus m. (c) The field at which the one-loop potential of the vacuum has its maximum, and the field beyond which it lies below the vacuum value, in units of the vacuum value. (d) The temperature at which the fermions have lost half of their mass, against the coupling, for three ratios m_s/m. Data: `data/medium_history.csv`, `data/matter_saturation.csv`, `data/vacuum_history.csv`, `data/thermal_history.csv`; programs: `deflation history`, `deflation paired`.

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4. Temperature: the fermion mass melts earlier and earlier

At high temperature the expectation value of the Vorton-Scalar vanishes and the fermions are massless (S4, Sec. 6). S4 gives two closed estimates of the temperature at which this happens and computes, in the Hartree approximation, the temperatures $T_{1/2}$ and T_0 at which the expectation value has fallen to one half and to (nearly) zero. All are of the order of the four-dimensional vacuum value $v_{VS} = m/g_{VS}$, which falls as $e^{-\tau_d/2}$ while the masses stay fixed. Along the clock (check `def-H7`):

m_s/m	g_{VS}	C	$T_{\rm high}/m$	$T_{1/2}/m$	T_0/m	melting defined
0.30	0.30	1.00	0.939	0.857	0.941	yes
0.30	0.40	1.78	0.749	0.691	0.751	yes

m_s/m	g_{VS}	C	T_{high}/m	$T_{1/2}/m$	T_0/m	melting defined
0.30	0.53	3.16	0.617	0.573	0.618	yes
0.30	0.71	5.62	0.528	0.491	0.528	yes
0.30	0.95	10.00	0.471	0.435	0.471	yes
0.30	1.26	17.8	0.436	0.400	0.436	yes
0.30	1.69	31.6	0.415	—	—	no
0.30	2.25	56.2	0.402	—	—	no
0.30	3.00	100.0	0.395	—	—	no
0.30	4.00	178	0.391	—	—	no
1.0	0.30	0.0900	2.605	2.325	2.627	yes
1.0	0.40	0.160	1.967	1.776	1.991	yes
1.0	0.53	0.285	1.493	1.368	1.519	yes
1.0	0.71	0.506	1.143	1.065	1.170	yes
1.0	0.95	0.900	0.888	0.842	0.914	yes
1.0	1.26	1.60	0.705	0.679	0.726	yes
1.0	1.69	2.85	0.577	0.561	0.587	yes
1.0	2.25	5.06	0.490	0.476	0.490	yes
1.0	3.00	9.00	0.434	0.413	0.434	yes
1.0	4.00	16.0	0.399	0.366	0.399	yes
3.0	0.30	0.0100	4.809	4.536	4.899	yes
3.0	0.40	0.0178	3.609	3.492	3.719	yes
3.0	0.53	0.0316	2.710	2.707	2.842	yes
3.0	0.71	0.0562	2.037	2.115	2.190	yes
3.0	0.95	0.1000	1.535	1.668	1.706	yes
3.0	1.26	0.178	1.160	1.329	1.346	yes
3.0	1.69	0.316	0.881	1.070	1.075	yes
3.0	2.25	0.562	0.676	0.870	0.870	yes
3.0	3.00	1.000	0.527	0.713	0.713	yes
3.0	4.00	1.78	0.420	0.590	0.590	yes

In units of the constant fermion mass, $T_{1/2}$ falls monotonically as the coupling grows, for each of the three mass ratios. **Matter of a fixed temperature melts at a definite moment of the deflation.**

A limit of the calculation, stated plainly. For $m_s/m = 0.3$ no melting temperature can be given from $g_{VS} = 1.69$ ($C = 32$) on: the last four rows of that block. There the one-loop potential of the vacuum turns over so close to the vacuum value (Sec. 5) that the Hartree potential ceases to exist at 1.05 times the vacuum value before the expectation value has fallen. The program marks these moments as not defined rather than report a number. (An earlier version of the program reported the artefact of the routine as a temperature; the error is recorded in the provenance RP9, item R-E14.)

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5. The vacuum along the clock

5.1 The barrier moves in

The one-loop potential of the Vorton-Scalar in vacuum contains the negative contribution of the fermion loop, which grows with the fourth power of the field and eventually wins: beyond some

field the potential lies below its value in the vacuum (S4, Sec. 5.3). The larger C, the smaller that field (check `def-H6`):

C	field at the maximum $/v_{VS}$	field beyond which the potential is below its vacuum value $/v_{VS}$
0.490	252	324
1.01	18.9	24.2
9.89	1.55	1.78
20.3	1.25	1.37

At $C = 1$ the potential falls below its vacuum value only beyond 24.7 times the vacuum value; at $C = 10$ already at 1.77 times. As the deflation proceeds, the barrier between the vacuum of the model and the region where the one-loop potential is unbounded below moves in towards the vacuum.

What this does and does not mean. The one-loop potential is not reliable at fields far from the vacuum value, and a potential that falls at large field signals that the effective theory needs completion there, not that the vacuum decays. What the result does establish is a trend: every statement of the model that rests on the vacuum of the Vorton-Scalar becomes less secure with time, and beyond C of order ten the turning point is within a factor of two of the vacuum value. The order of this event relative to the binding of the pair depends on m_s/m alone and is given in RS1, Sec. 4.2: for $m_s/m < 4.9$ the potential has fallen below its vacuum value at three times the vacuum value *before* two fermions bind.

5.2 A pair condensate in the vacuum

S4, Sec. 7, asks whether the pair can condense without a medium, and finds from the ladder equation with a momentum cutoff that this requires α_{VS} above a critical value of order one. With the cutoff $100\,m$ the critical values are between 2.13 and 2.93 for the six mass ratios of RS1, Sec. 4.2. On the deflating spacetime α_{VS} reaches such a value at a definite moment — but at that moment the coupling is strong, the one-boson exchange is no longer a controlled approximation, and by Sec. 5.1 the vacuum of the model is no longer protected. The statement of S4 stands, with one addition: a pair condensate in the vacuum is not reached by the deflation within the range of validity of the model.

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6. What has been established, and what has not

Established.

1. Along the deflation clock the solution of the gap equation is a minimum of the CJT potential at every moment examined; its condensation energy, critical temperature, thermodynamic ratios, Ginzburg–Landau coefficients and coherence lengths are given in Sec. 2, with their weak-coupling laws and the moments at which these fail.
2. Matter of fixed density becomes abnormal at a critical value of C that the deflation reaches at a definite time (Sec. 3.1); matter is self-bound for $C > 3.7011$ and more strongly bound later (Sec. 3.2).
3. The melting temperature of the fermion mass falls along the clock (Sec. 4).
4. The barrier of the one-loop vacuum potential moves towards the vacuum value (Sec. 5.1).

Not established.

1. *Strong coupling.* Every history ends in a regime that the two-loop truncation does not control. The numbers at the last moments of each table indicate the trend; they are not predictions.

2. *Kinetics.* The sections give sequences of equilibrium states. Only for the paired state has the ability to follow been examined (RS1, Sec. 7). How a gas reaches the self-bound state, and whether the abnormal state forms by nucleation, are open.
3. *The vacuum beyond one loop.* Sec. 5.1 uses the one-loop potential outside its range of reliability, as S4 does, and draws only the conclusion stated there.
4. *The neglected term.* The shift of the boson masses by $\mathrm{gsum}^2/4$ is neglected throughout (Sec. 1). It would matter only for a rate of deflation comparable with a particle mass.

Appendix A. Complete output of the programs

A.1 `deflation paired`

```
PROJECT spacetimes, Revision_Deflation: the results of the CJT effective action along the
 ↪  deflation clock
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation paired
7 of 7 checks passed

[PASS] def-P1  (numerical) the paired state along the clock (13 moments, two particles): at every
 ↪  moment the solution of the gap equation is a minimum of the CJT potential (Newton's method on
 ↪  the potential reproduces the iterated gap function to 7.7e-12; the minimum along the ray lies
 ↪  at s = 1 to 4.6e-7; the smallest eigenvalue of the second variation is positive), the critical
 ↪  temperature from the potential equals that of the linearized gap equation to 2.6e-12, and the
 ↪  condensation energy is within 15 per cent of -nu N(0) gap^2/4 wherever the gap is below 0.1
 ↪  E_F. The condensation energy deepens monotonically, in weak coupling with twice the
 ↪  logarithmic slope of the gap (to within 0.1 per cent), and the coherence length of the pair
 ↪  shrinks: for the first particle from 25 to 1.0 in units of 1/k_F. At the last 5 moments of
 ↪  that particle the gap exceeds 0.1 E_F, and at the last one the coherence length has fallen to
 ↪  the distance 1/k_F between the particles: the deflation carries the paired state from
 ↪  overlapping pairs towards bound pairs
[PASS] def-P2  (numerical) thermodynamics along the clock: while the critical temperature rises
 ↪  from 6.34e-3 to 1.04e-1 E_F (g_VS = 2.6 to 3.4), the ratio gap(0)/T_c stays within 0.019 of
 ↪  the weak-coupling value 1.7639 and the jump of the heat capacity within 0.048 of 1.4261: both
 ↪  ratios are universal while the coupling is weak, and they move only slightly as the gap
 ↪  approaches 0.2 E_F at the last moment
[PASS] def-P3  (numerical) Ginzburg-Landau energy along the clock (three moments, T_c = 6.3e-3,
 ↪  3.3e-2, 1.0e-1 E_F): while T_c is below 0.05 E_F the coefficients from the CJT potential agree
 ↪  with the weak-coupling formulas to 1.0 per cent; at the third moment, T_c = 0.10 E_F, they
 ↪  differ by up to 7.9 per cent, the state leaving the weak-coupling regime. The quadratic
 ↪  coefficient keeps its slope 4 N(0), the quartic and the gradient coefficients fall as 1/T_c^2,
 ↪  and the coherence length, sqrt(7 zeta(3)/(48 pi^2)) v_F/T_c = 0.1333 v_F/T_c in weak coupling
 ↪  (computed: 0.1333, 0.1340, 0.1390), shrinks from 80 to 5.1 (units 1/m). Vortex cores shrink as
 ↪  the deflation proceeds
[PASS] def-P4  (numerical) the space of the order parameter on the deflating spacetimes (frame
 ↪  components along x0, x5, x6, x7; both spacetimes): every unitary state, whichever its
 ↪  direction, has all eight gaps equal and the potential -2 (units of N(0) gap^2), with four flat
 ↪  directions (the Goldstone modes) and four directions of positive curvature; the half-paired
 ↪  state has four gaps closed and the potential -1. The potential does not distinguish the
 ↪  deflating axes from x0: in this sector the deflation selects no direction of the order
 ↪  parameter
[PASS] def-P5  (numerical) the test of the CJT method on the zero-dimensional model, executed
 ↪  again: the two-loop truncation, the one used for the models, has a solution for every coupling
 ↪  from 0.01 to 30 and stays within 10.7 per cent of the exact propagator
[PASS] def-P6  (numerical) the moment of binding from the gap equation: the linearized gap
 ↪  equation in vacuum reaches the eigenvalue 1 at D_VS = 1.7347, 1.6902, 1.6810 for m_s/m = 0.3,
 ↪  0.1, 0.03, approaching the value 1.67981 of the Schroedinger equation as the binding becomes
 ↪  non-relativistic (relative differences 3.3e-2, 6.2e-3, 6.9e-4)
```

```
[PASS] def-P7  (numerical) self-bound matter along the clock: once C has passed 3.7011 the lowest
 ↪  energy per particle lies below m, and it falls monotonically as the deflation proceeds: from
 ↪  -0.0192 m at C = 4 to -0.405 m at C = 29.6. At every one of these moments the matter of lowest
 ↪  energy is in the abnormal state (fermion mass zero inside the matter, largest value of m*/m in
 ↪  the table 0.0e0), at a Fermi momentum that falls from 0.98 m to 0.59 m. The exchange energies
 ↪  of the radial and of the phase mode at that density are 0.44 and 0.51 per cent of the
 ↪  mean-field energy at C = 4, and 0.90 and 1.22 per cent at C = 29.6 (m_s/m = 0.3): the mean
 ↪  field remains a good approximation along the whole history
[PASS] ALL  (numerical) every check above passed
```

A.2 deflation history

```
PROJECT spacetimes, Revision_Deflation: checks of the history of the model along the deflation
 ↪  clock (couplings that grow as exp(gsum x4))
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation history
8 of 8 checks passed

[PASS] def-H1  (numerical) thresholds of the model, computed again: a pair of fermions is bound if
 ↪  D_VS > 1.67981; matter is self-bound if C > 3.7011 (3 pi^2/8 = 3.7011) or, with the
 ↪  negative-energy states, C > 5.9218 (3 pi^2/5 = 5.9218). On a deflating spacetime D_VS and C
 ↪  grow as exp(gsum x4), so each threshold is passed at a definite x4
[PASS] def-H2  (numerical) along the clock, wherever the gap is below 0.05 E_F the critical
 ↪  temperature is 0.567 times the gap (largest deviation 0.0020); the gap of the vector channel
 ↪  is at every moment at least as large as the gap with the radial mode alone, and that at least
 ↪  as large as the gap of the two-form channel; the gap never decreases as the deflation proceeds
 ↪  (four sets of m_s/m and k_F/m, 34 moments each)
[PASS] def-H3  (numerical) growth of the gap: for a weakly coupled state the gap is proportional
 ↪  to exp(-1/lambda), where lambda = lambda_S + lambda_P is the sum of the pairing strengths of
 ↪  the radial mode and of the phase mode in the vector channel, and lambda grows as exp(tau_d),
 ↪  which gives d ln(gap)/d tau_d = 1/lambda; in the computed histories lambda d ln(gap)/d tau_d
 ↪  lies between 1.00 and 1.03 wherever 1e-6 < gap/E_F < 0.02: the gap grows much faster than the
 ↪  couplings, by one e-folding in the time 1/(gsum/lambda)
[PASS] def-H4  (numerical) the moment at which a pair becomes bound: just below D_c the scattering
 ↪  length is large and negative (-134.0/m_s at 0.99 D_c), just above large and positive
 ↪  (135.9/m_s at 1.01 D_c); the binding energy starts as a power of the distance from the
 ↪  threshold with the exponents 2.01 (between 2 and 4 per cent above D_c) and 2.01 (between 4 and
 ↪  8 per cent), that is quadratically: E_b ~ (exp(tau_pair) - 1)^2 m_s^2/m. Because E_b vanishes
 ↪  at the threshold, the pair cannot follow the deflation adiabatically at first, however slow
 ↪  the deflation is
[PASS] def-H5  (numerical) matter of fixed density loses the fermion mass (the Vorton-Scalar loses
 ↪  its expectation value in the matter) when C passes a critical value that falls with the
 ↪  density: k_F = 0.2 m: C = 151.892 (2.46 times pi^2 m^2/(4 k_F^2)); k_F = 0.5 m: C = 12.971
 ↪  (1.31 times pi^2 m^2/(4 k_F^2)); k_F = 1 m: C = 2.556 (1.04 times pi^2 m^2/(4 k_F^2)); k_F = 2
 ↪  m: C = 0.617 (1.00 times pi^2 m^2/(4 k_F^2)). The estimate C = pi^2 m^2/(4 k_F^2) of the main
 ↪  project is reached at high density, where the transition happens at weak coupling; at low
 ↪  density the critical C is larger. On a deflating spacetime every piece of matter becomes
 ↪  abnormal at a definite x4, the denser the earlier
[PASS] def-H6  (numerical) the field beyond which the one-loop potential of the Vorton-Scalar lies
 ↪  below its vacuum value decreases monotonically as C grows: 24.7 v_VS at C = 1, 1.77 v_VS at C
 ↪  = 10. As the deflation proceeds the barrier that protects the vacuum moves in towards the
 ↪  vacuum value
[PASS] def-H7  (numerical) the temperature T_half at which the fermions lose half of their mass,
 ↪  in units of the (constant) fermion mass, falls monotonically as g_VS grows, for m_s/m = 0.3, 1
 ↪  and 3 (26 of the 30 moments examined; it is of the order of the four-dimensional vacuum value
 ↪  m/g_VS, which falls as exp(-gsum x4/2)): matter of a fixed temperature melts at a definite x4.
 ↪  No melting temperature can be given for m_s/m = 0.3 from g_VS = 1.69 (C = 32) on; there the
 ↪  Hartree potential ceases to exist at 1.05 v_VS before the expectation value has fallen,
 ↪  because the one-loop potential turns over too close to the vacuum value
```

```
[PASS] def-H8  (numerical) order of events: the pair of fermions becomes bound before the one-loop
 ↪  potential of the Vorton-Scalar has fallen below its vacuum value at a field of 3 v_VS only for
 ↪  the mass ratios m_s/m = 10, 30 of the six ratios 0.1, 0.3, 1, 3, 10, 30 examined (the
 ↪  condition is 4 pi D_c m/m_s < 4.27, i.e. m_s/m > 4.9): for a light radial mode the deflation
 ↪  destabilizes the vacuum first
[PASS] ALL  (numerical) every check above passed
```

Appendix B. Files

file	content	written by
`code/deflation-rs/src/cmd_paired.rs`	the paired state along the clock from the CJT potential of the main project	—
`code/deflation-rs/src/history.rs`	matter, vacuum and temperature along the clock	—
`data/cjt_history.csv`	Sec. 2.1	`deflation paired`
`data/cjt_thermo_history.csv`, `data/cjt_gl_history.csv`	Secs. 2.2 and 2.3	`deflation paired`
`data/order_parameter_states.csv`	Sec. 2.4	`deflation paired`
`data/cjt_toy.csv`, `data/cjt_threshold.csv`	Sec. 1 (test of the method) and Sec. 2.5	`deflation paired`
`data/matter_saturation.csv`	Sec. 3.2	`deflation paired`
`data/medium_history.csv`	Sec. 3.1	`deflation history`
`data/thermal_history.csv`	Sec. 4	`deflation history`
`data/vacuum_history.csv`, `data/events.csv`	Sec. 5	`deflation history`
`data/paired_checks.txt`, `data/history_checks.txt` (and `.csv`)	Appendix A	the two commands
`figures/fig_cjt_history`, `figures/fig_matter_history` (`.pdf`, `.png`)	the two figures	`make_figures.py`